

THE THEORY OF



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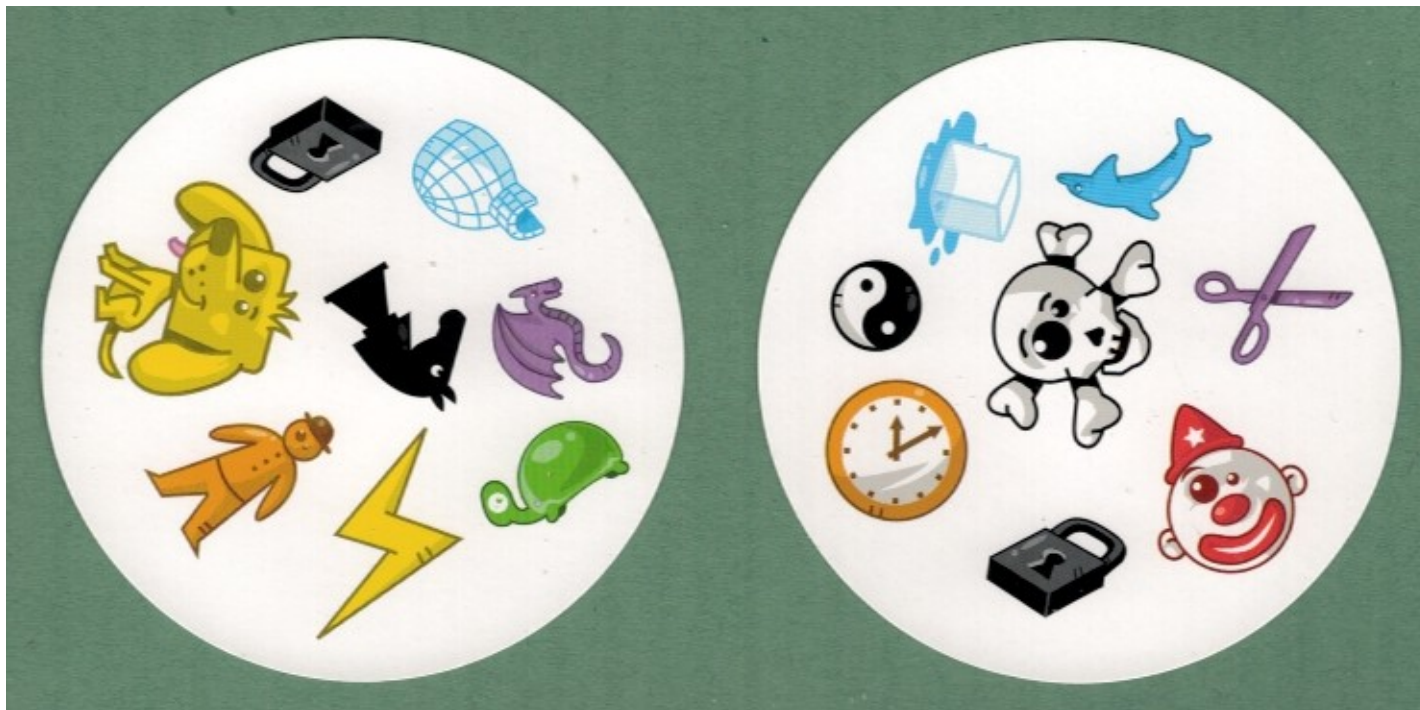
Spot the shared symbol . . .



Spot the shared symbol . . .



Spot the shared symbol . . .





In standard *Dobble*, there are ...

- 8 symbols on each card
- 55 cards
- 57 different symbols altogether

Main Rule: Any two cards share **exactly one** symbol.



QUESTION:

If you didn't have a *Dobble* set, how would you make one?

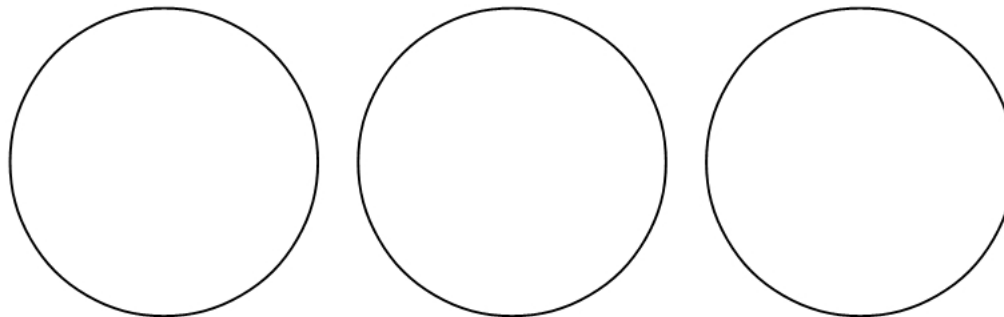
How would you know which symbols to put on which cards ...
so that **any two cards share exactly one symbol**?

Start with a **baby version** of *Dobble*.

Just 3 different symbols . . .



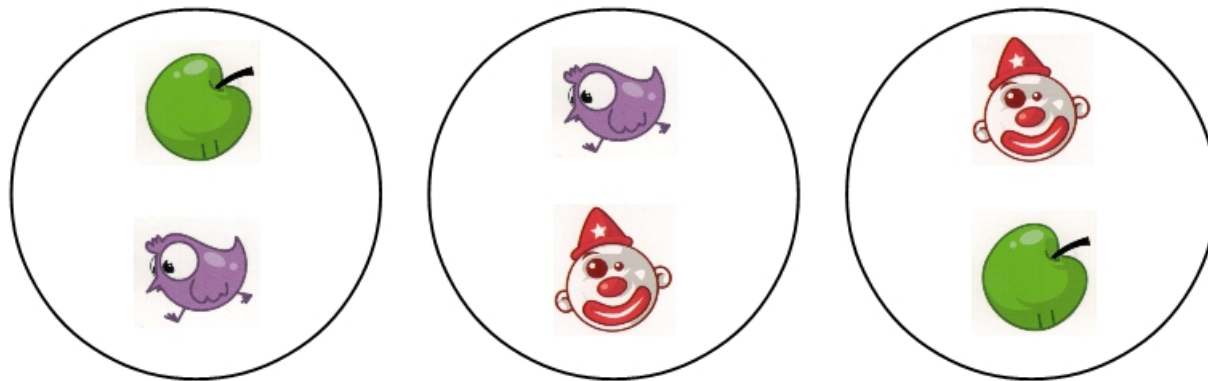
. . . and 3 cards:



Just **2** symbols on each card. How would you place them?

(Remember, any two cards must share exactly one symbol.)

Answer:



Let's call this *Level 2 Dobble* (2 symbols on each card).

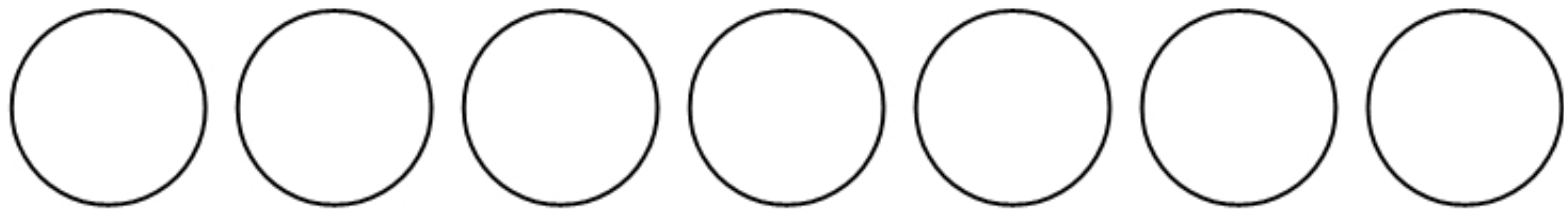
'Ordinary' Dobble is *Level 8 Dobble* (8 symbols per card).

Level 3 Dobble

Now choose a set of 7 symbols. They can be anything: e.g.

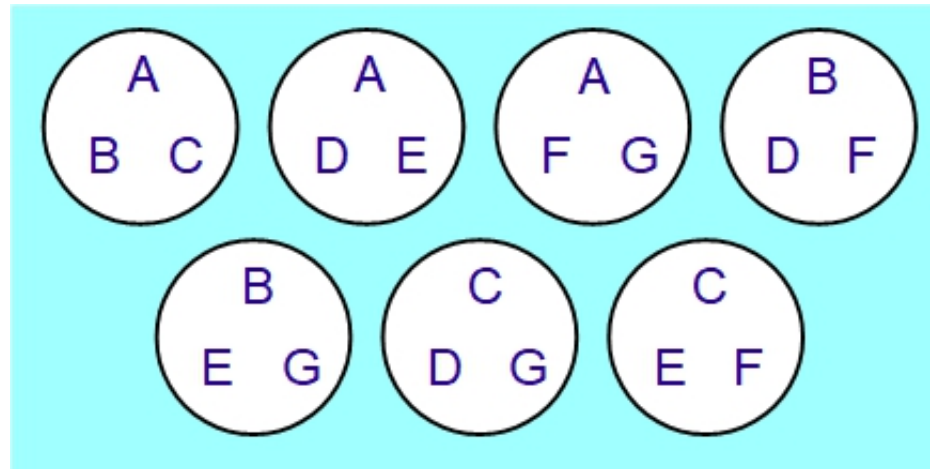
A B C D E F G 1 2 3 4 5 6 7 ★ ↑ ◇ ⊙ # ♭ ♯

Challenge: Arrange these on 7 cards, with 3 symbols per card.



Remember the rule: Any two cards must share just one symbol.

Level 3 Dobble: SOLUTION



Notice that ...

- every card has 3 symbols,
- any two cards share exactly one symbol,
- every symbol appears on 3 cards,

- any two symbols share exactly one card.

Let's say a Dobble set is complete if this last fact is true.

To try at home:

❄ Construct a complete **level 4** Dobble set
(4 symbols per card, 13 symbols, 13 cards).

❄ **HARDER:** Construct a complete **level 5** Dobble set
(5 symbols per card, 21 symbols, 21 cards).

Complete Dobble sets:

Level	Number of symbols/cards
1	1
2	3
3	7
4	13
5	21

Can you **spot a pattern** here? More on this in **handout sheet**.

Surprising connection! Dobble and geometry

In a complete Dobble set . . .

- any two cards have exactly one symbol in common,
- any two symbols have exactly one card in common.

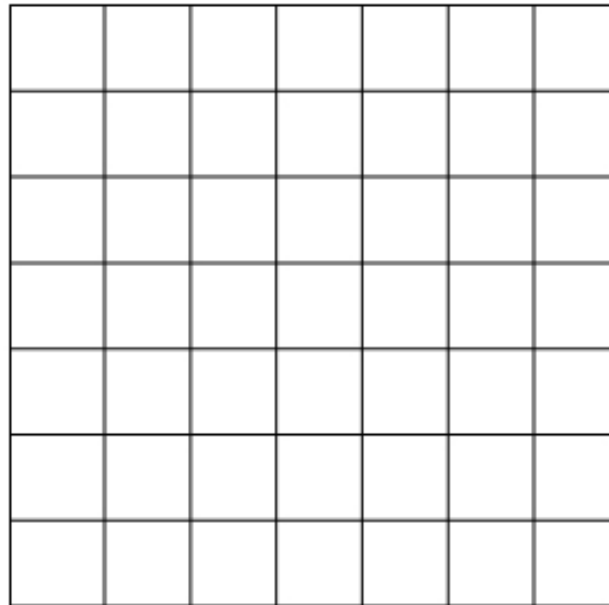
In geometry . . .

- any two points have exactly one line in common,
- ? any two lines have exactly one point in common ?
(Not quite true . . . but nearly!)

So perhaps some special kind of geometry can show us a way to make complete Dobble sets?

And perhaps our knowledge of geometry can then tell us some interesting things about Dobble?

A simple 'computer game' geometry (7×7)



Left and right edges 'join up'. So do top and bottom edges.

'Points' in this world are just squares.

What kinds of 'lines' are there?

(Some are in directions given by knight's moves. )

Every 'point' has 8 'lines' through it



IDEA: Match up points with cards, and lines with symbols.

Making a Level 8 Dobble set: Part 1

In our new geometry, there are 56 lines (7 in each of 8 directions). Label each line with some symbol.

For each of the 49 points, look at the 8 lines through it, and look at their labels. Create a card containing these 8 symbols.

That already gives us a Dobble set with 49 cards, 56 symbols.

Making a Level 8 Dobble set: Part 2

We can now squeeze in a few more cards.

For each of the 8 line directions, add a special **direction card**, containing the symbols for all 7 lines in that direction.

Finally, any two direction cards will need a symbol in common. So choose one last symbol to add to each direction card.

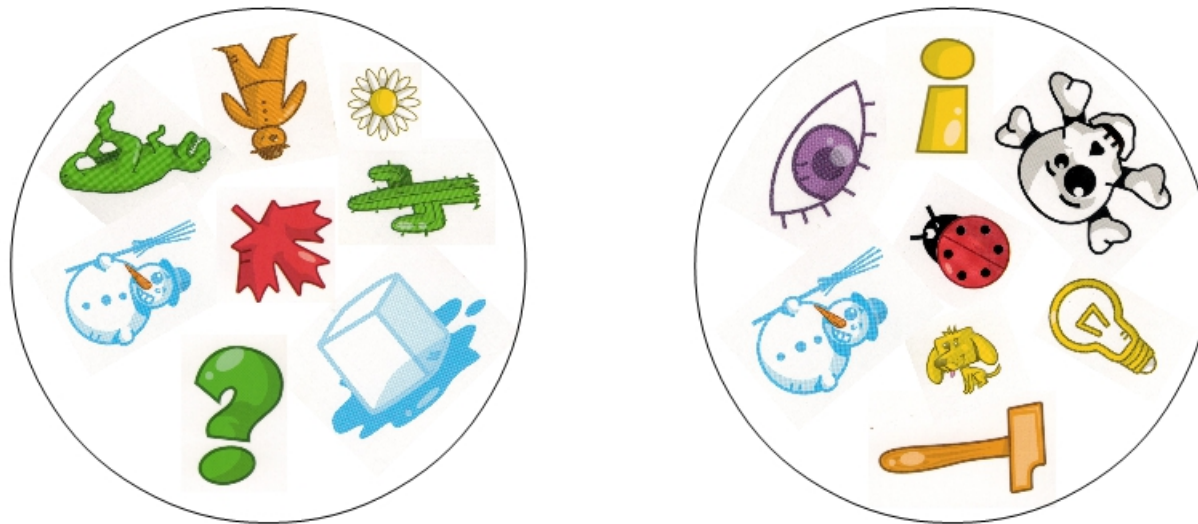


This actually gives us a **complete Level 8 Dobble set**.
(**57 cards, 57 symbols.**)

A bizarre omission

We've seen how to construct a **complete** Level 8 Dobble set with **57 cards, 57 symbols**.

BUT ordinary Dobble has only **55 cards!!** The missing two are:

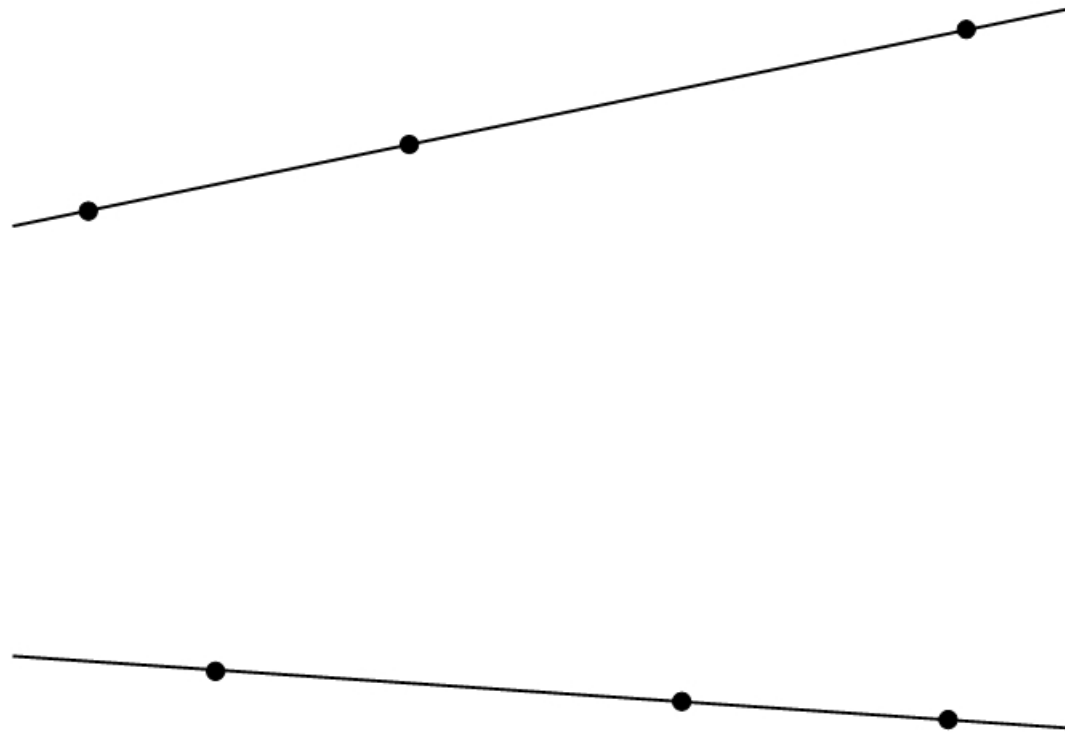


No idea why the manufacturers didn't include these!

Translating geometry facts into Dobble facts

For example, **Pappus's Theorem**.

(Named after *Pappus of Alexandria*, c. 300 AD.)

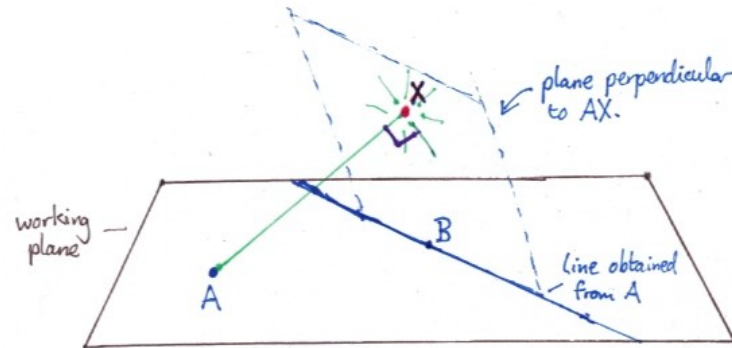


This works in *Dobble* too!

And finally (a bit harder) . . .

In **geometry**, there's a way of pairing up **points** with **lines**.

Need to think in 3D. Choose a point X **above** the plane we're working in.



Starting from a point A in our working plane, take the plane through X perpendicular to AX. This intersects the working plane in a line, which we'll call **the line obtained from A**.

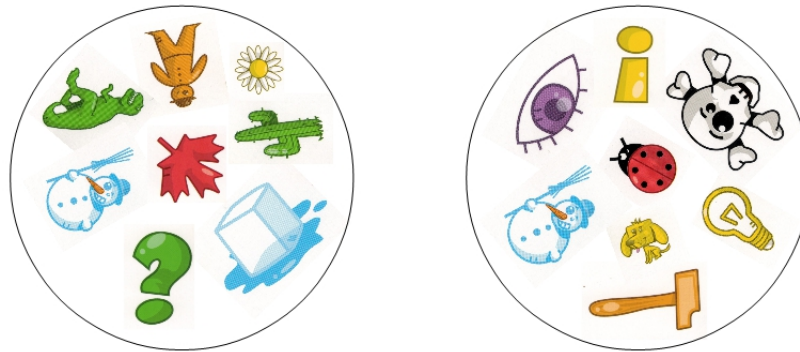
Cool fact: If B is on the line obtained from A, then also A is on the line obtained from B. (Because both of these just mean 'AXB is a right angle'.)

Again, all of this translates very neatly into **Dobble**.

Conclusion



Dobble has an amazing amount of mathematical structure.
(Shame about those two missing cards ...)



Thanks for listening!!